APPENDIX 1

A

Capstone Project Synopsis

On

FileMind – AI-Powered Document Intelligence System

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APPENDIX 2

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CHAPTER 1: INTRODUCTION

1.1 BRIEF OF THE PROJECT

In modern academic, professional, and commercial ecosystems, the volume of digital documents generated daily is expanding exponentially. Most of these documents are stored in Portable Document Format (PDF) due to its cross-platform layout fidelity and universal support. However, this ubiquity has created a profound operational crisis: individuals and organizations routinely amass vast archives of unstructured PDFs characterized by uninformative filenames (such as *doc23e3y49.pdf*, *scan0042.pdf*, or machine-generated export labels), fragmented folder structures, rampant document duplication, and an absence of unified organization. Conventional document storage systems—ranging from native operating system file directories to standard cloud storage drives—treat PDFs as passive, opaque binary blobs. While these systems excel at basic bit-level file persistence and streaming, they possess no semantic comprehension of the text, structure, or real-world meaning of the information contained inside the pages. Consequently, knowledge workers and students are forced to manually inspect, rename, and sort dozens or hundreds of files one by one, squandering valuable time and frequently failing to locate critical records.

To resolve these fundamental limitations, **FileMind** is conceived and engineered as an autonomous, AI-powered document intelligence system. FileMind eliminates the burdensome clerical overhead of manual classification, taxonomy tagging, and file renaming. Instead of requiring human intervention to organize incoming files, the system autonomously analyzes uploaded PDF documents and understands their semantic nature, contextual domain, and key data attributes directly from their internal content.

The core understanding engine of FileMind operates on Google's advanced multimodal artificial intelligence model, **Gemini 3.8 Flash**. When a PDF document is ingested into the system, the AI pipeline performs a holistic analysis that goes beyond simple optical character recognition. It automatically derives a definitive **document category** (e.g., Invoice, Receipt, Academic Transcript, Legal Contract, or Bank Statement), generates relevant descriptive **domain tags** (e.g., *tax*, *audit*, *tuition*, *vendor*), extracts standardized **key-value metadata** (such as invoice identifiers, currency, billing parties, dates, and net payable amounts), and captures clean **page-wise text extractions**. For instance, if an invoice bearing the obscure filename *doc23e3y49.pdf* is submitted, FileMind immediately detects that it is a vendor billing document, identifies invoice code *INV-9842-X*, extracts monetary totals, and indexes the text with zero human prompting.

Data security, multi-tenant isolation, and institutional privacy are architectural cornerstones of FileMind. All uploaded documents are stored in private, cryptographically segregated object storage paths using **Supabase Storage**. User identity and authentication are strictly regulated via **Supabase Auth**. At the database layer, PostgreSQL Row-Level Security (RLS) guarantees that document records, metadata, page texts, and retrieval indexes remain strictly isolated to the authenticated owner. Unauthorized cross-tenant access is categorically prevented by database-enforced policies rather than discretionary application logic.

The AI processing workflow follows a deterministic, state-machine lifecycle: **Upload** → **Pending** → **AI Processing** → **Schema Validation** → **Structured Persistence** → **Completed**. Robust runtime validation using **Zod** ensures that all AI-generated structured data satisfies rigid type constraints, coercing raw outputs and stripping invalid keys. Crucially, page integrity checks enforce that page numbering is fully contiguous (1, 2, ..., N) and matches the verified document page count before committing data.

Beyond document classification and structured parsing, FileMind incorporates an advanced **intelligent hybrid retrieval engine** (Phase 1B). Rather than relying purely on rudimentary lexical substring matches, the system transforms validated page-wise extracted text into high-dimensional mathematical representations using **Gemini Embedding 2**, producing 768-dimensional dense vectors. These embeddings are stored and indexed directly within PostgreSQL utilizing the **pgvector** extension configured with a **Hierarchical Navigable Small World (HNSW)** cosine index. To deliver maximum retrieval precision, FileMind implements a multi-channel hybrid search architecture combining dense semantic retrieval, PostgreSQL full-text search (GIN tsvector tokens), and deterministic exact identifier matching, fused through **Reciprocal Rank Fusion (RRF, $k = 60$)**.

Users can query their repository using natural language expressions (such as *"ABC company invoices from last quarter"*) or exact alphanumeric reference codes (such as *"INV-9842-X"*). Complex generative question-answering and Retrieval-Augmented Generation (RAG / "Ask This Document") represent the planned **Phase 1C** roadmap. FileMind transforms passive digital archives into structured, searchable, and intelligent knowledge repositories.

1.2 TECHNOLOGIES USED

The architecture of FileMind leverages an enterprise-grade technology stack chosen for security, mathematical retrieval performance, and operational reliability, summarized in Table 1.1 below.

Component / Layer	Technology / Tool	Functional Role in FileMind
Frontend Application	Next.js 16 (App Router) React 19 & TypeScript	Full-stack web application, responsive dashboard UI, encrypted PDF reader viewport.
Authentication	Supabase Auth	User lifecycle management, JWT session cookies, tenant identity verification.
Object Storage	Supabase Storage	Private PDF storage buckets, isolated cryptographic paths, short-lived signed URLs.
Relational Database	PostgreSQL / Supabase	Transactional metadata persistence, Row-Level Security (RLS), atomic RPC functions.
AI Understanding	Google Gemini 3.8 Flash (@google/genai SDK)	Multimodal document parsing, automatic categorization, tagging, and text extraction.
Structured Validation	Zod Runtime Schema	Validation, normalization, and typing of AI outputs; page sequence integrity checks.
Vector Embeddings	Gemini Embedding 2 (768 Dimensions)	High-dimensional semantic representation of chunked text for similarity search.
Vector Engine & Index	PostgreSQL pgvector (HNSW Index)	Sub-linear approximate nearest neighbor vector search with cosine distance metric.
Hybrid Search Engine	Reciprocal Rank Fusion (RRF, $k = 60$ + GIN FTS)	Fused ranking combining dense semantic search, lexical full-text, and exact match boost.
Deployment Environment	Vercel Cloud Platform	Global Edge CDN, serverless API Route Handlers, background worker execution.

Table 1.1: Core Architectural Technology Stack of FileMind.

CHAPTER 2: PROPOSED METHODOLOGY

The proposed methodology for FileMind encompasses a secure, multi-stage pipeline designed to transition raw PDF files from unorganized byte streams into rigorously validated, searchable, and intelligent data entities. The system architecture is structured across clear operational stages from initial user authentication to high-speed hybrid retrieval.

1. User Authentication & Session Establishment

The workflow initiates with secure user authentication managed by Supabase Auth. Upon successful login, cryptographic JSON Web Tokens (JWT) are issued and stored within secure, HTTP-only cookies. Every downstream API request derives user identity strictly from this authenticated server session. Client-supplied user identifiers are disregarded to prevent spoofing or impersonation.

2. Secure PDF Upload & Pre-flight Ingestion

FileMind supports both single and bulk document uploads. To ensure system stability and compliance with cloud infrastructure limits, a strict file size ceiling of **49 MB per individual PDF** is enforced. Crucially, this 49 MB constraint applies to each distinct PDF file independently, enabling users to select and submit large batches of documents simultaneously without exceeding batch limits. Before committing files to storage, the server performs pre-flight validation: verifying binary magic bytes (%PDF-1.x), validating MIME types, and sanitizing filenames against directory path traversal (../) and script injection vectors.

3. Isolated Object Persistence & Metadata Initialization

Validated documents are persisted into private Supabase Storage buckets under cryptographically isolated folder paths structured as {user_id}/{document_id}/{filename}. Concurrently, an initial database record is created in the public.documents table with an initial lifecycle status of pending. This stage enforces user storage quotas and tracks document file attributes before AI invocation.

4. Multimodal AI Document Understanding (Phase 1A)

The AI understanding worker executes server-side using Google's gemini-3.8-flash via the Gemini Files API. The model analyzes the complete visual and textual structure of the PDF across all pages. Under strict system instructions, the model extracts a high-level document category, contextual tags, generic normalized key-value metadata items, page counts, scanned document flags, and complete page-by-page extracted text. The model output is strictly constrained to a standardized JSON schema. Crucially, runtime parsing using **Zod** validates the extracted payload, coercing types and verifying that page numbers are contiguous and completely match the physical page count.

5. Processing Security, Lease Fencing & Recovery

To guarantee transactional integrity during asynchronous execution, FileMind incorporates an atomic lease fencing protocol. The server worker acquires an exclusive processing lock via the PostgreSQL RPC acquire_document_processing, which issues a fresh UUID processing_token and timestamps the lease. If processing exceeds a 10-minute timeout threshold or an environment crash occurs, the lease expires. Stale workers attempting to commit results are fenced out because their token no longer matches the database state. Furthermore, a dedicated recovery endpoint (POST /api/documents/[id]/retry) allows pending or stalled documents to be securely re-scheduled.

6. Structured Page-wise Text Storage

Upon successful validation and fencing checks, the atomic RPC finalize_document_processing commits the extracted metadata and inserts page records into public.document_pages. Each page entry maintains an exact mapping between the parent document_id, page_number, and validated extracted_text.

7. Deterministic Text Chunking (Phase 1B)

To transition page-wise text into granular, searchable retrieval units, the system implements a deterministic, boundary-aware chunker. The chunker processes sequentially ordered page texts, targeting segments of approximately **3,200 characters (~700–900 tokens)** with a bounded overlap of **500 characters (~100–150 tokens)**. It respects paragraph boundaries and preserves specialized alphanumeric identifiers (such as invoice codes, PAN, GSTIN, and dates) to prevent fragmenting critical tokens. Each chunk is assigned a contiguous 0-indexed chunk_index, tracks its source span (page_start to page_end), and computes a deterministic SHA-256 content_hash.

8. High-Dimensional Embedding Generation

Each text chunk is converted into a 768-dimensional dense vector using **Gemini Embedding 2**. Chunk texts are structured with semantic task prefixes (title: {doc_title} | text: {chunk_text}) to optimize representation density. Embedding calls are throttled through a bounded concurrency queue (maximum 5 concurrent requests) with exponential backoff and randomized jitter to handle transient API rate limits. Generated vectors are validated to ensure exactly 768 finite floating-point dimensions before database ingestion.

9. Vector Persistence & HNSW Indexing

Validated chunks and vectors are transactionally committed to the public.document_chunks table through the atomic RPC finalize_document_indexing. The table is indexed with a **Hierarchical Navigable Small World (HNSW)** cosine index (USING hnsw (embedding vector_cosine_ops)), which enables sub-linear approximate nearest neighbor search. Simultaneously, a generated fts_tokens tsvector column is indexed using a PostgreSQL Generalized Inverted Index (GIN) to empower rapid lexical keyword matching.

10. Multi-Channel Hybrid Search & Reciprocal Rank Fusion

When a user executes a search query, FileMind triggers a multi-channel retrieval engine. The user query is converted into a 768-dimensional vector to retrieve the top 30 semantic candidates via match_document_chunks. In parallel, the query text is evaluated against the lexical index via search_document_chunks_lexical to retrieve the top 30 full-text candidates. The candidates are merged using **Reciprocal Rank Fusion (RRF)** with constant $k = 60$:

$$\textbf{RRF Score} = \left[\frac{1}{60 + \text{Rank}_{\text{semantic}}}\right] + \left[\frac{1}{60 + \text{Rank}_{\text{lexical}}}\right] + \textbf{Exact Match Boost}$$

A deterministic exact match boost (+0.05) is awarded to chunks containing exact alphanumeric identifier matches, guaranteeing that precise codes (such as reference numbers or transaction IDs) surface at the absolute top of results.

11. Contextual Snippet Extraction & Deep Viewer Linking

Search results return bounded text snippets (maximum 750 characters) contextually anchored around the matched query terms. In the client interface, each result displays the document title, category, score, and source page range. Clicking a result opens the integrated PDF viewer automatically deep-linked to the corresponding page (#page=X).

12. Architectural Scope Boundary

Phase 1B concludes with complete hybrid retrieval and deep-linked document inspection. Higher-level generative capabilities—such as conversational Q&A;, LLM-generated answers, and automated synthesis—are strictly designated as Phase 1C future scope.

CHAPTER 3: APPLICATIONS AND SCOPE

3.1 PRACTICAL APPLICATIONS

The automated intelligence, structured classification, and hybrid retrieval capabilities of FileMind address critical operational bottlenecks across a wide variety of practical domains:

1. **Personal Document Management:** Individuals routinely accumulate hundreds of critical PDFs, including academic degrees, tax filings, insurance policies, medical records, and utility receipts. FileMind eliminates chaotic naming, allowing users to instantly locate records via natural inquiries like *"life insurance premium 2025"* or policy numbers.
2. **Educational Institutions & Universities:** Academic administrations manage immense volumes of student records, admission dossiers, examination transcripts, and departmental circulars. FileMind facilitates rapid archiving and instant retrieval of student records by roll number or semester without tedious manual indexing.
3. **Corporate & Enterprise Operations:** Businesses handle continuous inflows of vendor invoices, purchase orders, non-disclosure agreements, and employee contracts. The system streamlines accounts payable and legal archiving by categorizing incoming documents and extracting payment amounts, billing entities, and due dates autonomously.
4. **Accounting & Chartered Accountancy (CA) Firms:** Auditing and accounting professionals navigate thousands of client receipts, bank statements, and GST returns. FileMind's exact identifier boosting ensures that specific invoice numbers (e.g. *INV-9842-X*) or tax registration numbers are retrieved instantaneously with 100% precision.
5. **Institutional Archives & Digital Libraries:** Historical document repositories, research paper archives, and public records can be ingested in bulk and made semantically discoverable without requiring manual metadata curation for every volume.
6. **Administrative & Clerical Workflow Optimization:** FileMind significantly reduces bureaucratic delay and manual filing errors in government and administrative offices by automating document intake, taxonomy assignment, and departmental routing.

3.2 FUTURE SCOPE

While the current implementation of FileMind delivers a robust, verified hybrid retrieval and document understanding platform (Phases 0, 1A, and 1B), several advanced research and development vectors are planned for future phases:

1. **Retrieval-Augmented Generation (RAG) & Document Q&A; (Phase 1C):** Implementing conversational intelligence enabling users to "Ask This Document" complex questions, with generative answers synthesised directly from retrieved chunks.
2. **Page-Grounded Synthesis with Exact Citations:** Augmenting generative responses with verbatim sentence-level citations and direct visual bounding boxes on original PDF pages to eliminate AI hallucinations.
3. **Multi-Document Cross-Comparison & Summarization:** Providing comparative analysis across multiple related documents (e.g., comparing quarterly financial statements or vendor contract clauses side by side).
4. **Intelligent Duplicate & Version Detection:** Developing semantic deduplication algorithms to detect identical or revised documents across large archives, tracking version history automatically.
5. **Proactive Expiry & Renewal Reminders:** Implementing automated temporal tracking to alert users ahead of document expiration dates (e.g., passport renewals, warranty lapses, or lease expirations).
6. **Multi-Format Ingestion Support:** Extending the ingestion pipeline beyond PDFs to natively support Microsoft Word (DOCX), spreadsheets (XLSX), presentations (PPTX), and high-resolution scanned image formats (TIFF, PNG).
7. **Enterprise Workflow & ERP Integration:** Constructing pre-built webhooks and API connectors to synchronize with enterprise systems (SAP, Salesforce, QuickBooks) and cloud storage platforms (Google Drive, OneDrive).

Conclusion: FileMind demonstrates the efficacy of fusing state-of-the-art multimodal AI document understanding with pgvector-backed HNSW hybrid retrieval. By automating the classification, extraction, and indexing lifecycle, FileMind provides a resilient, production-ready foundation for next-generation intelligent document repositories.